Name
Date
Course

Adding and Subtracting Algebraic Fractions (Homework)

Combine the fractions below. **Answers**

(1) $\dfrac{3}{x^2+2x-8} + \dfrac{2}{x+4}$ $\boxed{\dfrac{2x-1}{(x-2)(x+4)}}$

(2) $\dfrac{4}{y} - \dfrac{1}{5y}$ $\boxed{\dfrac{19}{5y}}$

(3) $\dfrac{7}{b+6} + \dfrac{2}{b}$ $\boxed{\dfrac{9b+12}{b(b+6)}}$

(4) $\dfrac{3}{c^2-7c+10} + \dfrac{4}{c^2-25}$ $\boxed{\dfrac{7c+7}{(c-5)(c+5)(c-2)}}$

(5) $-\dfrac{8}{x} + \dfrac{6}{7}$ $\boxed{\dfrac{6x-56}{7x}}$

*Distribute the negative property when subtracting

(6) $\dfrac{3}{y+2} - \dfrac{5}{y-7}$ $\boxed{\dfrac{-2y-31}{(y+2)(y-7)}}$

(7) $\dfrac{7b}{b^2-b-12} + \dfrac{-10}{b-4}$ $\boxed{\dfrac{-3b-30}{(b-4)(b+3)}}$

(8) $\dfrac{2c}{5} + \dfrac{4}{3c}$ $\boxed{\dfrac{6c^2+20}{15c}}$

(9) $\dfrac{1}{7x} - \dfrac{6x+2}{5}$ $\boxed{\dfrac{-42x^2-14x+5}{35x}}$

(10) $\dfrac{6}{2y-1} - \dfrac{7}{4y^2-2y}$ $\boxed{\dfrac{12y-7}{2y(2y-1)}}$

(11) $\dfrac{-3}{b+4} + \dfrac{6b}{5}$ $\boxed{\dfrac{6b^2+24b-15}{5(b+4)}}$

(12) $\dfrac{c-3}{c+3} + \dfrac{4}{c-2}$ $\boxed{\dfrac{c^2-c+18}{(c+3)(c-2)}}$

(13) $\dfrac{8}{x+5} - \dfrac{x-10}{x+4}$ $\boxed{\dfrac{-x^2+13x+82}{(x+5)(x+4)}}$

(14) $\dfrac{4x-1}{3x^3-3x} + \dfrac{-1}{3x^2-3x}$ $\boxed{\dfrac{3x-2}{3x(x-1)(x+1)}}$

You may write this in factored form also.
$(5y-7)(y-2)$

(15) $\dfrac{8y+7}{5y^2-17y+14}+\dfrac{3}{y}$ $\quad\boxed{\dfrac{23y^2-44y+42}{y(5y^2-17y+14)}}$

(16) $\dfrac{6b^2+5}{b^2+2b+1}-\dfrac{4}{b}$ $\quad\boxed{\dfrac{6b^3-4b^2-3b-4}{b(b^2+2b+1)}}$

(17) $\dfrac{1}{6c}+\dfrac{2}{c-3}+7$ $\quad\boxed{\dfrac{42c^2-113c-3}{6c(c-3)}}$

(18) $\dfrac{2x-7}{6x^2+36x}-\dfrac{x+2}{5x-5}$ $\quad\boxed{\dfrac{-6x^3-38x^2-117x+35}{30x(x+6)(x-1)}}$

May also be written in expanded form.
$30x^3+150x^2-180x.$